

```
import joblib

from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler

# Assume model and scaler already trained

joblib.dump(model, "music_genre_model.pkl")
joblib.dump(scaler, "scaler.pkl")

print("Model Saved Successfully")
```

Interface Pipelines

```
import joblib
import pandas as pd

# Load saved files
model = joblib.load("music_genre_model.pkl")
scaler = joblib.load("scaler.pkl")

def predict_genre(features):

    df = pd.DataFrame([features])

    # Same preprocessing used during training
    scaled_data = scaler.transform(df)

    prediction = model.predict(scaled_data)[0]
```

```
confidence = model.predict_proba(scaled_data).max()
```

```
return prediction, confidence
```

```
sample_song = {  
    "tempo": 120,  
    "energy": 0.85,  
    "danceability": 0.72,  
    "loudness": -5.3,  
    "acousticness": 0.15  
}
```

```
genre, confidence = predict_genre(sample_song)
```

```
print("Genre:", genre)
```

```
print("Confidence:", round(confidence*100,2), "%")
```

app.py

```
import streamlit as st
```

```
import pandas as pd
```

```
import joblib
```

```
model = joblib.load("music_genre_model.pkl")
```

```
scaler = joblib.load("scaler.pkl")
```

```
st.title("🎵 Music Genre Classification App")
```

```
uploaded_file = st.file_uploader(  
    "Upload CSV File",  
    type=["csv"]  
)
```

```
if uploaded_file:
```

```
    df = pd.read_csv(uploaded_file)
```

```
    scaled_data = scaler.transform(df)
```

```
    predictions = model.predict(scaled_data)
```

```
    df["Predicted Genre"] = predictions
```

```
    st.subheader("Prediction Results")
```

```
    st.dataframe(df)
```

Example CSV

```
tempo,energy,danceability,loudness,acousticness  
120,0.8,0.75,-5.4,0.12  
90,0.4,0.30,-12.0,0.85  
150,0.9,0.80,-3.0,0.05
```

```
st.header("🎧 Single Song Prediction")
```

```
tempo = st.number_input("Tempo", value=120.0)
```

```
energy = st.number_input("Energy", value=0.8)
```

```
danceability = st.number_input("Danceability", value=0.7)
```

```
loudness = st.number_input("Loudness", value=-5.0)
acousticness = st.number_input("Acousticness", value=0.2)
```

```
song = pd.DataFrame([{
    "tempo": tempo,
    "energy": energy,
    "danceability": danceability,
    "loudness": loudness,
    "acousticness": acousticness
}])
```

```
scaled_song = scaler.transform(song)
```

```
prediction = model.predict(scaled_song)[0]
```

```
probabilities = model.predict_proba(scaled_song)[0]
```

```
confidence = max(probabilities)
```

```
st.success(
    f"Predicted Genre: {prediction}"
)
```

```
st.metric(
    "Confidence Score",
    f"{confidence*100:.2f}%"
)
```

```
st.subheader("Summary")
```

```
genre_counts = df["Predicted Genre"].value_counts()
```

```
st.bar_chart(genre_counts)
```